Appendices

Sets, functions, fields, complex numbers and polynomials

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A Sets

Motivation. The language of sets, unions, intersections and equivalence relations used throughout the notes.

A **set** is a collection of objects (**elements**); $x \in A$ means x is an element of A. $A = B$ if they have the same elements; $A \subseteq B$ (A is a **subset** of B) if every element of A is in B; $A \subsetneq B$ is a **proper** subset. The **empty set** $\emptyset$ has no elements. Sets are described by listing, $\{1, 2, 3\}$, or by a property, $\{x \in \mathbb{R} : 1 < x < 2\}$.

Example 1

$A = \{x \in \mathbb{R} : 1 < x < 2\}$, the reals between 1 and 2; $\{1, 2\} = \{2, 1\}$ (order and repetition do not matter).

The **union** $A \cup B = \{x : x \in A \text{ or } x \in B\}$, the **intersection** $A \cap B = \{x : x \in A \text{ and } x \in B\}$; A, B are **disjoint** if $A \cap B = \emptyset$. For a family $\{A_\alpha : \alpha \in \Lambda\}$, $\bigcup_\alpha A_\alpha$ and $\bigcap_\alpha A_\alpha$ are defined likewise.

Example 2

$A = \{1, 3, 5\}$, $B = \{1, 5, 7, 8\}$: $A \cup B = \{1, 3, 5, 7, 8\}$, $A \cap B = \{1, 5\}$.

Example 3

For $\alpha > 1$ let $A_\alpha = \{x \in \mathbb{R} : -\frac{1}{\alpha} \leq x \leq 1 + \alpha\}$. Then $\bigcup_{\alpha>1} A_\alpha = \{x : -1 < x\}$ and $\bigcap_{\alpha>1} A_\alpha = \{x : 0 \leq x \leq 2\}$.

A **relation** on A is a set S of ordered pairs of elements of A; write $x \sim y$ if $(x, y) \in S$. It is an **equivalence relation** if it is **reflexive** ($x \sim x$), **symmetric** ($x \sim y \Rightarrow y \sim x$) and **transitive** ($x \sim y$, $y \sim z \Rightarrow x \sim z$). Examples: equality; congruence mod m on $\mathbb{Z}$; similarity and congruence of matrices; isomorphism of vector spaces.

B Functions

Motivation. One-to-one, onto and invertible functions, composition and restriction.

A **function** $f : A \to B$ assigns to each $x \in A$ a unique $f(x) \in B$; A is the **domain**, B the **codomain**, and $f(A) = \{f(x) : x \in A\}$ the **range**. For $S \subseteq A$, $f(S)$ is the **image**; for $T \subseteq B$, $f^{-1}(T) = \{x : f(x) \in T\}$ is the **preimage**. $f = g$ if $f(x) = g(x)$ for all x.

Example 1

$f : [-10, 10] \to \mathbb{R}$, $f(x) = x^2 + 1$: the range is $[1, 101]$, $f([2, 3]) = [5, 10]$, and $f^{-1}([5, 10]) = [-3, -2] \cup [2, 3]$.

f is **one-to-one** if $f(x) = f(y)$ implies $x = y$, and **onto** if $f(A) = B$. The **restriction** f_S of f to $S \subseteq A$ is f on S.

Example 2

$f : [-1, 1] \to [0, 1]$, $f(x) = x^2$ is onto but not one-to-one ($f(-1) = f(1)$); $f_{[0,1]}$ is one-to-one and onto.

The **composite** of $f : A \to B$ and $g : B \to C$ is $g \circ f : A \to C$, $(g \circ f)(x) = g(f(x))$; composition is associative but not commutative. f is **invertible** if some $g : B \to A$ has $g \circ f = \mathsf{I}_A$ and $f \circ g = \mathsf{I}_B$; then $g = f^{-1}$ is unique. Facts: f is invertible iff it is one-to-one and onto; $(f^{-1})^{-1} = f$; if f, g are invertible, so is $g \circ f$, with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Example 3

$f(x) = 3x + 1$ on $\mathbb{R}$ is one-to-one and onto, with $f^{-1}(x) = \frac{x-1}{3}$.

C Fields

Motivation. Scalars may come from any field: a set where one can add, subtract, multiply and divide by nonzero elements. The characteristic of a field (for example 2 in $\mathbb{Z}_2$) occasionally matters.

Definition (Field)

A **field** $\mathbb{F}$ is a set with operations $+$ and $\cdot$ such that for all $a, b, c \in \mathbb{F}$: (F1) $a + b = b + a$, $ab = ba$; (F2) $(a + b) + c = a + (b + c)$, $(ab)c = a(bc)$; (F3) there are distinct $0, 1$ with $0 + a = a$, $1 \cdot a = a$; (F4) each a has c with $a + c = 0$, and each $b \neq 0$ has d with $bd = 1$; (F5) $a(b + c) = ab + ac$.

Example 1

$\mathbb{R}$ is a field.

Example 2

$\mathbb{Q}$ is a field.

Example 3

$\{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a field: $(a + b\sqrt{2})^{-1} = \frac{a - b\sqrt{2}}{a^2 - 2b^2}$ ($a^2 \neq 2b^2$ for rationals not both zero).

Example 4

$\mathbb{Z}_2 = \{0, 1\}$ with $0 + 0 = 1 + 1 = 0$, $0 + 1 = 1$, $0 \cdot 0 = 0 \cdot 1 = 0$, $1 \cdot 1 = 1$ is a field.

Example 5

$\mathbb{N}$ and $\mathbb{Z}$ are not fields: (F4) fails (2 has no multiplicative inverse in $\mathbb{Z}$).

Theorem C.1 (Cancellation laws)

(a) If $a + b = c + b$, then $a = c$.
(b) If $ab = cb$ and $b \neq 0$, then $a = c$.

Proof.

(a) Add d with $b + d = 0$: $a = a + (b + d) = (a + b) + d = (c + b) + d = c$.

(b) Multiply by b^{-1} similarly. ■

Corollary

The elements 0, 1 of (F3) and the c, d of (F4) are unique.

Proof. If $0'$ also works, $0' + b = b = 0 + b$, so $0' = 0$ by (a); if $a + c = 0 = a + c'$, then $c = c'$ by (a). Similarly for 1 and d using (b). ■

Write $-a$ and b^{-1}; $a - b = a + (-b)$, $a/b = ab^{-1}$.

Theorem C.2

(a) $a \cdot 0 = 0$.
(b) $(-a)b = a(-b) = -(ab)$.
(c) $(-a)(-b) = ab$.

Proof.

(a) $0 + a \cdot 0 = a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0$; cancel.

(b) $ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$, so $(-a)b = -(ab)$; similarly $a(-b)$.

(c) By (b) twice, $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$. ■

Corollary

0 has no multiplicative inverse.

Proof. $0 \cdot d = 0 \neq 1$ by (a) and (F3). ■

The **characteristic** of $\mathbb{F}$ is the least $p \geq 1$ with $1 + \cdots + 1$ (p summands) $= 0$, or 0 if no such p exists. $\mathbb{Z}_2$ has characteristic 2; $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ have characteristic 0. In characteristic 2, $x + x = 0$ for all x, so 2 cannot be inverted; some results (e.g. Theorem 6.35) exclude this case. A nonzero characteristic is prime.

D Complex numbers

Motivation. $\mathbb{C}$ is the field in which every nonconstant polynomial has a zero. This is why complex scalars make eigenvalue theory complete.

Definition (Complex numbers)

A **complex number** is $z = a + bi$ with $a, b \in \mathbb{R}$ (real and imaginary parts); $(a+bi)+(c+di) = (a + c) + (b + d)i$ and $(a + bi)(c + di) = (ac - bd) + (bc + ad)i$. Then $i^2 = -1$, and $\mathbb{R} \subseteq \mathbb{C}$ via $a = a + 0i$.

Example 1

$z = 3 - 5i$, $w = 9 + 7i$: $z + w = 12 + 2i$, $zw = (27 + 35) + (-45 + 21)i = 62 - 24i$.

Example 2

$(-5+2i)(1-3i) = -5+15i+2i-6i^2 = 1+17i.$

Theorem D.1

$\mathbb{C}$ is a field.

Proof [Supplement]. (F1), (F2), (F5) are direct computations; $0 = 0+0i$, $1 = 1+0i$; $-(a+bi) = -a-bi$; and for $a+bi \neq 0$, $(a+bi)\frac{a-bi}{a^2+b^2} = 1$. ■

Definition (Conjugate)

$\overline{a+bi} = a-bi$.

Example 3

$\overline{-3+2i} = -3-2i$, $\overline{4-7i} = 4+7i$, $\bar{6} = 6$.

Theorem D.2

(a) $\bar{\bar{z}} = z$;
(b) $\overline{z+w} = \bar{z}+\bar{w}$;
(c) $\overline{zw} = \bar{z}\bar{w}$;
(d) $\overline{z/w} = \bar{z}/\bar{w}$ $(w \neq 0)$;
(e) $z \in \mathbb{R}$ iff $\bar{z} = z$.

Proof. Direct computation with $z = a+bi$, $w = c+di$; e.g. (c): $\overline{zw} = (ac-bd)-(bc+ad)i = (a-bi)(c-di)$. (d) follows from (c) applied to $(z/w)w = z$. ■

Definition (Absolute value)

$|a+bi| = \sqrt{a^2+b^2}$; then $z\bar{z} = |z|^2$, and $\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$.

Example 4

$\frac{1+4i}{3-2i} = \frac{(1+4i)(3+2i)}{9+4} = \frac{-5+14i}{13}$.

Theorem D.3

(a) $|zw| = |z||w|$;
(b) $|z/w| = |z|/|w|$ $(w \neq 0)$;
(c) $|z+w| \leq |z|+|w|$;
(d) $|z|-|w| \leq |z+w|$.

Proof.

(a) $|zw|^2 = zw\overline{zw} = z\bar{z}w\bar{w}$.

(b) From (a).

(c) $|z+w|^2 = |z|^2 + 2\,\mathrm{Re}(z\bar{w}) + |w|^2 \leq |z|^2 + 2|z||w| + |w|^2$, since $\mathrm{Re}\,u \leq |u|$.

(d) $|z| = |(z+w)+(-w)| \le |z+w|+|w|$. ■

Polar form: $z = |z|(\cos\theta + i\sin\theta) = |z|e^{i\theta}$, and zw multiplies absolute values and adds angles; $z^n = |z|^n e^{in\theta}$, so every complex number has n-th roots.

Theorem D.4 (Fundamental theorem of algebra)

Every $p(z) = a_n z^n + \cdots + a_0 \in \mathsf{P}(\mathbb{C})$ of degree $n \ge 1$ has a zero.

Proof.

Step 1 (a minimum exists). Let $\mu = \inf_z |p(z)|$. For $|z| = s$ large, $|p(z)| \ge |a_n|s^n - \sum_{k<n}|a_k|s^k > \mu + 1$, say for $s > R$. On the closed disk $|z| \le R$, the continuous function $|p|$ attains its minimum at some z_0 (compactness), and $|p(z_0)| = \mu$.

Step 2 (d'Alembert). Suppose $p(z_0) \neq 0$. Let $q(z) = p(z+z_0)/p(z_0)$, a polynomial of degree n with $q(0) = 1$ and $|q| \ge 1$ everywhere. Write $q(z) = 1 + b_k z^k + \cdots + b_n z^n$ with $b_k \neq 0$. Choose θ with $e^{ik\theta}b_k = -|b_k|$ (k-th roots exist). For $r > 0$ small with $r^k|b_k| < 1$,

$$|q(re^{i\theta})| \le 1 - r^k|b_k| + r^{k+1}\sum_{j>k}|b_j|r^{j-k-1} = 1 - r^k\Big(|b_k| - r\sum_{j>k}|b_j|r^{j-k-1}\Big) < 1$$

for r small enough, contradicting $|q| \ge 1$. So $p(z_0) = 0$. ■

Corollary

Every $p \in \mathsf{P}(\mathbb{C})$ of degree $n \ge 1$ factors as $p(z) = a_n(z-c_1)\cdots(z-c_n)$.

Proof. Induction on n: by Theorem D.4 and Corollary 1 of Theorem E.1, $p(z) = (z-c_1)q(z)$ with $\deg q = n-1$ and leading coefficient a_n. ■

A field in which every nonconstant polynomial splits is **algebraically closed**; $\mathbb{C}$ is, $\mathbb{R}$ is not.

E Polynomials

Motivation. Division with remainder, the Euclidean property "relatively prime $\Rightarrow q_1f_1 + q_2f_2 = 1$" and unique factorization into irreducibles are the polynomial facts used for eigenvalues, Cayley–Hamilton and decompositions of operators.

$f \mid g$ (f **divides** g) if $g = fq$ for some polynomial q.

Theorem E.1 (Division algorithm)

If $\deg f = n$ and $\deg g = m \ge 0$, there are unique q, r with $f = qg + r$ and $\deg r < m$.

Proof.

Step 1 (existence). Induction on n. If $n < m$, take $q = 0$, $r = f$. Otherwise, with leading coefficients a_n of f and b_m of g, $h = f - a_n b_m^{-1}x^{n-m}g$ has degree $< n$, so $h = q_1 g + r$ by induction, and $f = (q_1 + a_n b_m^{-1}x^{n-m})g + r$.

Step 2 (uniqueness). If $qg + r = q'g + r'$, then $(q-q')g = r' - r$ has degree $< m$, which forces $q = q'$ (otherwise the left side has degree $\ge m$) and then $r = r'$. ■

Corollary 1

For $\deg f > 0$ and $a \in \mathbb{F}$: $f(a) = 0$ iff $(x - a) \mid f$.

Proof. $f = q \cdot (x - a) + r$ with r constant; evaluating at a, $r = f(a)$. ■

Corollary 2

A polynomial of degree $n \geq 1$ has at most n distinct zeros.

Proof. Induction: if $f(a) = 0$, $f = (x - a)q$ with $\deg q = n - 1$, and every other zero of f is a zero of q (a field has no zero divisors). ■

Definition (Relatively prime)

Nonzero f, g are **relatively prime** if no polynomial of positive degree divides both. E.g. $x^2(x-1)$ and $(x-1)(x-2)$ are not; $x^2(x-1)$ and $(x-2)(x-3)$ are.

Theorem E.2

If f_1, f_2 are relatively prime, there are q_1, q_2 with $q_1 f_1 + q_2 f_2 = 1$.

Proof. Let $I = \{q_1 f_1 + q_2 f_2\}$, a set closed under addition and under multiplication by polynomials, containing nonzero elements. Let $d \in I$ be nonzero of least degree. By the division algorithm, $f_1 = qd + r$ with $\deg r < \deg d$ and $r = f_1 - qd \in I$; minimality forces $r = 0$, so $d \mid f_1$, and likewise $d \mid f_2$. As f_1, f_2 are relatively prime, $\deg d = 0$: $d = c \neq 0$, and $c^{-1}d = 1 \in I$. ■

Example 1

$f_1 = x^3 - x^2 + 1$ and $f_2 = (x-1)^2$ are relatively prime; indeed $(-x+2)f_1 + (x^2 - x - 1)f_2 = 1$.

Definition (Polynomials of operators)

For $f(x) = a_0 + a_1 x + \cdots + a_n x^n$, $f(\mathsf{T}) = a_0 \mathsf{I} + a_1 \mathsf{T} + \cdots + a_n \mathsf{T}^n$ and $f(A) = a_0 I + a_1 A + \cdots + a_n A^n$.

Example 2

$\mathsf{T}(a, b) = (2a + b, a - b)$ and $f(x) = x^2 + 2x - 3$: $\mathsf{T}^2(a, b) = (5a + b, a + 2b)$, so $f(\mathsf{T})(a, b) = (5a + b, a + 2b) + (4a + 2b, 2a - 2b) - (3a, 3b) = (6a + 3b, 3a - 3b)$.

Theorem E.3

(a) $f(\mathsf{T})$ is a linear operator.
(b) If β is a finite ordered basis and $A = [\mathsf{T}]_\beta$, then $[f(\mathsf{T})]_\beta = f(A)$.

Proof.

(a) Linear combinations of composites of linear maps are linear (Theorems 2.7, 2.9).

(b) $[\mathsf{T}^k]_\beta = A^k$ (Corollary to Theorem 2.11), and Theorem 2.8. ■

Theorem E.4

$f_1(\mathsf{T})f_2(\mathsf{T}) = f_2(\mathsf{T})f_1(\mathsf{T})$ and $f_1(A)f_2(A) = f_2(A)f_1(A)$.

Proof. Both equal $(f_1 f_2)(\mathsf{T})$, since powers of T commute and composition distributes over sums (Theorem 2.10). ■

Theorem E.5

If f_1, f_2 are relatively prime, there are q_1, q_2 with (a) $q_1(\mathsf{T})f_1(\mathsf{T}) + q_2(\mathsf{T})f_2(\mathsf{T}) = \mathsf{I}$; (b) $q_1(A)f_1(A) + q_2(A)f_2(A) = I$.

Proof. Apply $g \mapsto g(\mathsf{T})$, which respects sums and products, to the identity of Theorem E.2. ■

Definition (Monic, irreducible)

f is **monic** if its leading coefficient is 1; it is **irreducible** if $\deg f > 0$ and f is not a product of two polynomials of positive degree. (Irreducibility depends on $\mathbb{F}$: $x^2 + 1$ is irreducible over $\mathbb{R}$, not over $\mathbb{C}$.) Degree-one polynomials are irreducible.

Theorem E.6

If ϕ is irreducible and $\phi \nmid f$, then ϕ and f are relatively prime.

Proof. A common divisor g of positive degree divides ϕ, so $\phi = cg$ with c constant (irreducibility); then $\phi \mid f$, a contradiction. ■

Theorem E.7

Two distinct irreducible monic polynomials are relatively prime.

Proof. If $\phi_1 \mid \phi_2$, then $\phi_2 = c\phi_1$ (irreducible), and $c = 1$ (monic), so $\phi_1 = \phi_2$. Hence $\phi_1 \nmid \phi_2$; apply Theorem E.6. ■

Theorem E.8

If ϕ is irreducible and $\phi \mid fg$, then $\phi \mid f$ or $\phi \mid g$.

Proof. If $\phi \nmid f$, then $q_1\phi + q_2 f = 1$ (Theorems E.6, E.2), so $g = q_1\phi g + q_2 fg$ is divisible by ϕ. ■

Corollary

If irreducible monic ϕ divides $\phi_1 \cdots \phi_n$ (irreducible monic), then $\phi = \phi_i$ for some i.

Proof. Induction on n: $\phi \mid \phi_1 \cdots \phi_{n-1}$ (then induction) or $\phi \mid \phi_n$ (then $\phi = \phi_n$ by Theorem E.7). ■

Theorem E.9 (Unique factorization)

Every f of positive degree is uniquely $f = c\,\phi_1^{n_1} \cdots \phi_k^{n_k}$ with c a constant, ϕ_i distinct irreducible monic polynomials and $n_i \geq 1$.

Proof.

Step 1 (existence). Induction on $\deg f$. If f is irreducible, $f = c \cdot (c^{-1}f)$. Otherwise $f = gh$ with $0 < \deg g, \deg h < \deg f$; factor g and h by induction and collect equal factors.

Step 2 (uniqueness). c is the leading coefficient. If $\prod \phi_i^{n_i} = \prod \psi_j^{m_j}$, then ϕ_1 divides the right side, so $\phi_1 = \psi_j$ for some j (Corollary to Theorem E.8). Cancel one factor (no zero divisors) and induct on the total degree. ■

Theorem E.10

If $\mathbb{F}$ is infinite and $f(a) = g(a)$ for all $a \in \mathbb{F}$, then $f = g$.

Proof. $h = f - g$ vanishes at infinitely many points. If $\deg h \geq 1$, this contradicts Corollary 2 of Theorem E.1. So h is constant, and $h(a) = 0$ gives $h = 0$. ■

Over a finite field this fails: x^2 and x define the same function on $\mathbb{Z}_2$.